Linear Algebra: A Reference

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A compact reference for a proof-based first course in linear algebra, in the conventions of Friedberg, Insel & Spence. Statements are given without proof. Throughout, $\mathbb{F}$ denotes a field (usually $\mathbb{R}$ or $\mathbb{C}$), and all vector spaces are over $\mathbb{F}$ unless stated otherwise.

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1 Notation

$\mathbb{F}^n$	column vectors $(a_1, \ldots, a_n)^t$ with entries in $\mathbb{F}$
$\mathsf{M}_{m\times n}(\mathbb{F})$	$m \times n$ matrices over $\mathbb{F}$; A_{ij} is the (i,j) entry
$\mathsf{P}_n(\mathbb{F})$, $\mathsf{P}(\mathbb{F})$	polynomials of degree $\leq n$; all polynomials
$e_1, \ldots, e_n$	the standard basis of $\mathbb{F}^n$; δ_{ij} is the Kronecker delta
I_V, I_n	identity map on V; $n \times n$ identity matrix
$\operatorname{span}(S)$	set of all (finite) linear combinations of elements of S; $\operatorname{span}(\varnothing) = \{0\}$
$\mathcal{L}(V,W)$, $\mathcal{L}(V)$	linear maps $V \to W$; linear operators on V
$\mathsf{N}(\mathsf{T})$, $\mathsf{R}(\mathsf{T})$	null space (kernel) and range (image) of T
$\operatorname{nullity}(\mathsf{T})$, $\operatorname{rank}(\mathsf{T})$	$\dim \mathsf{N}(\mathsf{T})$ and $\dim \mathsf{R}(\mathsf{T})$
$[v]_\beta$	coordinate vector of v relative to an ordered basis β
$[\mathsf{T}]_\beta^\gamma$, $[\mathsf{T}]_\beta$	matrix of T relative to β (domain) and γ (codomain); $[\mathsf{T}]_\beta = [\mathsf{T}]_\beta^\beta$
L_A	left-multiplication map $x \mapsto Ax$, $\mathbb{F}^n \to \mathbb{F}^m$
A^t, A^*	transpose; conjugate transpose $(A^*)_{ij} = \overline{A_{ji}}$
$\det(A)$, $\operatorname{tr}(A)$	determinant; trace $\sum_i A_{ii}$
$W_1 \oplus W_2$	direct sum: $W_1 + W_2$ with $W_1 \cap W_2 = \{0\}$
V^*, T^t	dual space $\mathcal{L}(V,\mathbb{F})$; transpose (dual) map $g \mapsto g \circ \mathsf{T}$
V/W	quotient space, cosets $v + W$
E_λ, K_λ	eigenspace $\mathsf{N}(\mathsf{T} - \lambda\mathsf{I})$; generalized eigenspace
$f(t)$, $p(t)$	characteristic polynomial $\det(A - tI)$; minimal polynomial
$\langle x, y\rangle$, $\|x\|$	inner product; norm $\sqrt{\langle x, x\rangle}$
$S^\perp$	orthogonal complement $\{x : \langle x, y\rangle = 0 \text{ for all } y \in S\}$
T^*	adjoint of T: $\langle \mathsf{T}x, y\rangle = \langle x, \mathsf{T}^* y\rangle$

2 Vector Spaces

Definition 2.1 (vector space)**.** A set V with addition $V \times V \to V$ and scalar multiplication $\mathbb{F} \times V \to V$ such that for all $x, y, z \in V$ and $a, b \in \mathbb{F}$: (VS1) $x + y = y + x$; (VS2) $(x + y) + z = x + (y + z)$; (VS3) there is $0 \in V$ with $x + 0 = x$; (VS4) each x has y with $x + y = 0$; (VS5) $1x = x$; (VS6) $(ab)x = a(bx)$; (VS7) $a(x + y) = ax + ay$; (VS8) $(a + b)x = ax + bx$.

Standard examples: $\mathbb{F}^n$, $\mathsf{M}_{m\times n}(\mathbb{F})$, $\mathsf{P}_n(\mathbb{F})$, $\mathsf{P}(\mathbb{F})$, and the space $\mathcal{F}(S,\mathbb{F})$ of all functions from a set S to $\mathbb{F}$. Consequences of the axioms: the zero vector and additive inverses are unique, $0x = 0$, $a0 = 0$, and $(-a)x = -(ax)$.

2.1 Subspaces

Theorem 2.2 (subspace test)**.** *A subset $W \subseteq V$ is a subspace if and only if $0 \in W$, and $x + y \in W$ and $cx \in W$ whenever $x, y \in W$ and $c \in \mathbb{F}$.*

Intersections of subspaces are subspaces; unions generally are not. $W_1 \cup W_2$ is a subspace iff $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. The *sum* $W_1 + W_2 = \{w_1 + w_2\}$ is the smallest subspace containing both, and $V = W_1 \oplus W_2$ iff every $v \in V$ is *uniquely* $w_1 + w_2$.

2.2 Span, independence, bases

Definition 2.3. S is *linearly dependent* if $a_1u_1 + \cdots + a_nu_n = 0$ for some distinct $u_i \in S$ and scalars not all zero; otherwise *linearly independent*. A *basis* is a linearly independent set that spans V. The *dimension* $\dim V$ is the number of vectors in any basis.

Theorem 2.4 (replacement theorem)**.** *If $V = \operatorname{span}(G)$ with $|G| = n$ and $L \subseteq V$ is linearly independent with $|L| = m$, then $m \leq n$, and there is $H \subseteq G$ with $|H| = n - m$ such that $L \cup H$ spans V.*

Consequences, for $\dim V = n < \infty$:

- every basis has exactly n elements; any spanning set has $\geq n$ vectors and contains a basis;
- any linearly independent set has $\leq n$ vectors and extends to a basis;
- a set of exactly n vectors is a basis iff it spans iff it is linearly independent;
- $\beta = \{u_1, \ldots, u_n\}$ is a basis iff each $v \in V$ is uniquely $\sum a_iu_i$;
- a subspace W has $\dim W \leq n$, with equality iff $W = V$.

Dimension of a sum. For finite-dimensional subspaces, $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$. Common dimensions: $\dim \mathbb{F}^n = n$, $\dim \mathsf{M}_{m\times n}(\mathbb{F}) = mn$, $\dim \mathsf{P}_n(\mathbb{F}) = n + 1$; $\mathsf{P}(\mathbb{F})$ is infinite-dimensional.

2.3 Quotient spaces

For a subspace W, the cosets $v+W$ form a vector space V/W under $(v_1+W)+(v_2+W) = (v_1+v_2)+W$ and $a(v+W) = av+W$; $v_1+W = v_2+W$ iff $v_1-v_2 \in W$. If V is finite-dimensional, $\dim(V/W) = \dim V - \dim W$.

3 Linear Transformations

Definition 3.1 (linear map)**.** $\mathsf{T}\colon V \to W$ is linear if $\mathsf{T}(x + y) = \mathsf{T}x + \mathsf{T}y$ and $\mathsf{T}(cx) = c\,\mathsf{T}x$; equivalently $\mathsf{T}(cx+y) = c\mathsf{T}x+\mathsf{T}y$. Then $\mathsf{T}(0) = 0$. Its *null space* is $\mathsf{N}(\mathsf{T}) = \{x : \mathsf{T}x = 0\}$ and its *range* is $\mathsf{R}(\mathsf{T}) = \{\mathsf{T}x\}$; both are subspaces.

Theorem 3.2 (rank–nullity (dimension theorem))**.** *If V is finite-dimensional and $\mathsf{T}\colon V \to W$ is linear, then* $\operatorname{nullity}(\mathsf{T}) + \operatorname{rank}(\mathsf{T}) = \dim V$.

- T is one-to-one iff $\mathsf{N}(\mathsf{T}) = \{0\}$.
- If $\dim V = \dim W < \infty$, then T is one-to-one $\iff$ onto $\iff$ $\operatorname{rank}\mathsf{T} = \dim V$.
- If β is a basis of V, then $\mathsf{R}(\mathsf{T}) = \operatorname{span}(\mathsf{T}(\beta))$.
- A linear map is determined by its values on a basis, and any assignment of values on a basis extends uniquely to a linear map.

3.1 Matrix representations

For ordered bases $\beta = \{v_1, \dots, v_n\}$ of V and $\gamma = \{w_1, \dots, w_m\}$ of W, write $\mathsf{T}v_j = \sum_i a_{ij} w_i$. Then $[\mathsf{T}]_\beta^\gamma = (a_{ij}) \in \mathsf{M}_{m\times n}(\mathbb{F})$: *column j is* $[\mathsf{T}v_j]_\gamma$.

Theorem 3.3 (representation)**.**

$$[\mathsf{T}(v)]_\gamma = [\mathsf{T}]_\beta^\gamma\,[v]_\beta, \qquad [\mathsf{UT}]_\alpha^\gamma = [\mathsf{U}]_\beta^\gamma [\mathsf{T}]_\alpha^\beta, \qquad [\mathsf{T}+c\,\mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma + c\,[\mathsf{U}]_\beta^\gamma.$$

The map $\mathsf{T} \mapsto [\mathsf{T}]_\beta^\gamma$ is an isomorphism $\mathcal{L}(V,W) \cong \mathsf{M}_{m\times n}(\mathbb{F})$, so $\dim \mathcal{L}(V,W) = mn$.

3.2 Invertibility and isomorphism

T is invertible if there is U with $\mathsf{UT} = \mathsf{I}_V$ and $\mathsf{TU} = \mathsf{I}_W$; then T^{-1} is linear and unique. T is invertible iff it is one-to-one and onto, and in that case $[\mathsf{T}^{-1}]_\gamma^\beta = ([\mathsf{T}]_\beta^\gamma)^{-1}$. Finite-dimensional V and W are isomorphic iff $\dim V = \dim W$; in particular $V \cong \mathbb{F}^n$ via $v \mapsto [v]_\beta$.

3.3 Change of coordinates

Theorem 3.4 (change of basis)**.** *Let β, β' be ordered bases of V and $Q = [\mathsf{I}_V]_{\beta'}^\beta$ (its columns are the β-coordinates of the vectors of β'). Then Q is invertible, $[v]_\beta = Q[v]_{\beta'}$, and for $\mathsf{T} \in \mathcal{L}(V)$,*

$$[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_\beta\, Q.$$

Matrices A, B are *similar* if $B = Q^{-1}AQ$ for some invertible Q. Similar matrices share rank, determinant, trace, characteristic polynomial, eigenvalues (with multiplicities), and minimal polynomial.

3.4 Dual spaces

$V^* = \mathcal{L}(V, \mathbb{F})$. If $\beta = \{x_1, \dots, x_n\}$ is a basis of V, the *dual basis* $\beta^* = \{f_1, \dots, f_n\}$ is defined by $f_i(x_j) = \delta_{ij}$, and every $f \in V^*$ satisfies $f = \sum_i f(x_i) f_i$. So $\dim V^* = \dim V$ when finite. The transpose $\mathsf{T}^t \colon W^* \to V^*$, $\mathsf{T}^t(g) = g\mathsf{T}$, satisfies $[\mathsf{T}^t]_{\gamma^*}^{\beta^*} = ([\mathsf{T}]_\beta^\gamma)^t$. The map $x \mapsto \hat{x}$, $\hat{x}(f) = f(x)$, is a natural isomorphism $V \cong V^{**}$ for finite-dimensional V.

4 Matrices and Linear Systems

4.1 Elementary operations and rank

The three elementary row (column) operations are: swap two rows; multiply a row by a nonzero scalar; add a multiple of one row to another. Each equals left (right) multiplication by an invertible *elementary matrix*.

Theorem 4.1 (rank)**.** $\operatorname{rank}(A) = \operatorname{rank}(L_A)$ *equals the dimension of the column space, which equals the dimension of the row space. Multiplying by invertible matrices preserves rank, so elementary operations preserve rank, and* $\operatorname{rank}(A) = \operatorname{rank}(A^t)$*. Moreover* $\operatorname{rank}(AB) \le \min\{\operatorname{rank} A, \operatorname{rank} B\}$*.*

Every matrix of rank r can be reduced by elementary operations to $\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$, and every invertible matrix is a product of elementary matrices. To invert A, row reduce $(A \mid I_n)$ to $(I_n \mid A^{-1})$.

4.2 Systems $Ax = b$

- The solution set K_H of $Ax = 0$ is $\mathsf{N}(L_A)$, of dimension $n - \operatorname{rank} A$.
- If s is any solution of $Ax = b$, the full solution set is $s + K_H$.
- $Ax = b$ is consistent iff $\operatorname{rank}(A) = \operatorname{rank}(A \mid b)$.
- If A is $n \times n$, then $Ax = b$ has a unique solution for every b iff A is invertible.
- Gaussian elimination: row reduce $(A \mid b)$ to reduced row echelon form; the RREF of a matrix is unique.

The invertible matrix theorem. For $A \in \mathsf{M}_{n\times n}(\mathbb{F})$ the following are equivalent: A is invertible; $\operatorname{rank} A = n$; $\mathsf{N}(L_A) = \{0\}$; L_A is onto; the columns (or rows) of A form a basis of $\mathbb{F}^n$; $Ax = b$ has a unique solution for every b; A is a product of elementary matrices; A row reduces to I_n; $\det A \neq 0$; 0 is not an eigenvalue of A.

5 Determinants

Theorem 5.1 (characterization)**.** $\det\colon \mathsf{M}_{n\times n}(\mathbb{F}) \to \mathbb{F}$ *is the unique function that is linear in each row, is zero whenever two rows are equal (alternating), and satisfies* $\det(I_n) = 1$*. Explicitly,*

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, A_{1\sigma(1)} A_{2\sigma(2)} \cdots A_{n\sigma(n)},$$

and by cofactor expansion along row i*,* $\det(A) = \sum_j (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij})$*, where* $\tilde{A}_{ij}$ *deletes row* i *and column* j*.*

- Swapping two rows negates det; scaling a row by k scales det by k; adding a multiple of a row leaves det unchanged.
- $\det(AB) = \det(A)\det(B)$, $\det(A^t) = \det(A)$, and $\det(A^{-1}) = \det(A)^{-1}$.

- A is invertible iff $\det A \neq 0$. The determinant of a triangular matrix is the product of its diagonal entries.
- Cramer's rule: if $\det A \neq 0$, the solution of $Ax = b$ is $x_k = \det(M_k)/\det(A)$, where M_k replaces column k of A by b.
- Over $\mathbb{R}$, $|\det A|$ is the factor by which L_A scales n-dimensional volume; the sign records orientation.

6 Eigenvalues and Diagonalization

Definition 6.1. $\lambda \in \mathbb{F}$ is an *eigenvalue* of $\mathsf{T} \in \mathcal{L}(V)$ if $\mathsf{T}v = \lambda v$ for some $v \neq 0$, an *eigenvector*. The *characteristic polynomial* of $A \in \mathsf{M}_{n\times n}(\mathbb{F})$ is $f(t) = \det(A - tI_n)$, of degree n with leading coefficient $(-1)^n$; that of T is that of any $[\mathsf{T}]_\beta$. The *eigenspace* is $E_\lambda = \mathsf{N}(\mathsf{T} - \lambda\mathsf{I})$.

- λ is an eigenvalue iff $f(\lambda) = 0$. The *algebraic multiplicity* of λ is its multiplicity as a root of f; the *geometric multiplicity* is $\dim E_\lambda$, and $1 \leq \dim E_\lambda \leq$ algebraic multiplicity.
- Eigenvectors for distinct eigenvalues are linearly independent.
- If f splits, $\det A = \prod \lambda_i$ and $\operatorname{tr} A = \sum \lambda_i$ (with multiplicity).

Theorem 6.2 (diagonalizability)**.** T *is* diagonalizable *if V has a basis of eigenvectors of T, i.e. $[\mathsf{T}]_\beta$ is diagonal for some β. This holds iff (1) f splits over $\mathbb{F}$ and (2) for each eigenvalue, the geometric multiplicity equals the algebraic multiplicity; equivalently, $V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}$. In particular, n distinct eigenvalues suffice. Then $A = QDQ^{-1}$, where the columns of Q are eigenvectors and D is diagonal, and $A^m = QD^mQ^{-1}$.*

6.1 Invariant subspaces and Cayley–Hamilton

A subspace W is T-*invariant* if $\mathsf{T}(W) \subseteq W$; then the characteristic polynomial of $\mathsf{T}|_W$ divides that of T. The T-*cyclic subspace* generated by $x \neq 0$ is $W = \operatorname{span}\{x, \mathsf{T}x, \mathsf{T}^2x, \dots\}$; if $\dim W = k$, then $\{x, \dots, \mathsf{T}^{k-1}x\}$ is a basis and $\mathsf{T}^k x = -\sum_{i<k} a_i \mathsf{T}^i x$ gives the characteristic polynomial $(-1)^k(a_0 + a_1 t + \cdots + a_{k-1}t^{k-1} + t^k)$ of $\mathsf{T}|_W$.

Theorem 6.3 (Cayley–Hamilton)**.** *If f is the characteristic polynomial of T, then $f(\mathsf{T}) = \mathsf{T}_0$, the zero operator. Likewise $f(A) = 0$ for matrices.*

6.2 Minimal polynomial

The *minimal polynomial* p of T is the unique monic polynomial of least positive degree with $p(\mathsf{T}) = \mathsf{T}_0$.

- $g(\mathsf{T}) = \mathsf{T}_0$ iff p divides g; in particular $p \mid f$.
- p and f have the same roots: the eigenvalues of T.
- T is diagonalizable iff p splits over $\mathbb{F}$ with no repeated roots, i.e. $p(t) = (t-\lambda_1)\cdots(t-\lambda_k)$ for distinct λ_i.

7 Canonical Forms

Definition 7.1 (generalized eigenvectors)**.** $x \neq 0$ is a *generalized eigenvector* for λ if $(\mathsf{T}-\lambda \mathsf{I})^p x = 0$ for some $p \geq 1$. The *generalized eigenspace* $K_\lambda = \{x : (\mathsf{T}-\lambda\mathsf{I})^p x = 0 \text{ for some } p\}$ is T-invariant and contains E_λ.

Theorem 7.2 (Jordan canonical form)**.** *If the characteristic polynomial of* T *splits, with distinct eigenvalues* $\lambda_1, \dots, \lambda_k$ *of multiplicities* $m_1, \dots, m_k$*, then* $\dim K_{\lambda_i} = m_i$*,* $K_{\lambda_i} = \mathsf{N}\big((\mathsf{T}-\lambda_i \mathsf{I})^{m_i}\big)$*, and* $V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k}$*. There is an ordered basis (a union of cycles of generalized eigenvectors) in which*

$$[\mathsf{T}]_\beta = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_r \end{pmatrix}, \qquad J = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix}.$$

This Jordan form *is unique up to the order of the blocks; two matrices with split characteristic polynomials are similar iff they have the same Jordan form.*

- The number of Jordan blocks for λ equals $\dim E_\lambda$.
- The number of blocks for λ of size $\geq j$ equals $\operatorname{rank}\big((\mathsf{T}-\lambda\mathsf{I})^{j-1}\big) - \operatorname{rank}\big((\mathsf{T}-\lambda\mathsf{I})^{j}\big)$ (dot diagrams).
- In the minimal polynomial, the exponent of $(t-\lambda)$ is the size of the largest Jordan block for λ.

Over fields where f need not split, the *rational canonical form* plays the same role, built from companion matrices of powers of irreducible factors of f.

8 Inner Product Spaces

Definition 8.1 (inner product)**.** On V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, an inner product $\langle \cdot, \cdot \rangle$ satisfies, for all x, y, z and c: $\langle x+z, y\rangle = \langle x, y\rangle + \langle z, y\rangle$; $\langle cx, y\rangle = c\langle x, y\rangle$; $\overline{\langle x, y\rangle} = \langle y, x\rangle$; $\langle x, x\rangle > 0$ for $x \neq 0$. It is therefore conjugate-linear in the second argument: $\langle x, cy\rangle = \bar{c}\langle x, y\rangle$. The standard inner product on $\mathbb{F}^n$ is $\langle x, y\rangle = \sum_i x_i \overline{y_i} = y^* x$; on $\mathsf{M}_{n\times n}(\mathbb{F})$ it is $\langle A, B\rangle = \operatorname{tr}(B^* A)$.

Theorem 8.2 (basic inequalities)**.** *For* $\|x\| = \sqrt{\langle x, x\rangle}$*: Cauchy–Schwarz* $|\langle x, y\rangle| \leq \|x\|\|y\|$*; triangle inequality* $\|x+y\| \leq \|x\| + \|y\|$*; Pythagoras* $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ *when* $x \perp y$*.*

8.1 Orthonormal bases and Gram–Schmidt

A set is *orthonormal* if its vectors are pairwise orthogonal unit vectors; orthogonal sets of nonzero vectors are linearly independent. If $\beta = \{v_1, \dots, v_n\}$ is an orthonormal basis, then

$$x = \sum_{i=1}^{n} \langle x, v_i\rangle v_i, \qquad \|x\|^2 = \sum_i |\langle x, v_i\rangle|^2, \qquad ([\mathsf{T}]_\beta)_{ij} = \langle \mathsf{T} v_j, v_i\rangle.$$

Theorem 8.3 (Gram–Schmidt). *If $\{w_1, \dots, w_n\}$ is linearly independent, define $v_1 = w_1$ and*

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} v_j \qquad (2 \le k \le n).$$

Then $\{v_1, \dots, v_n\}$ is orthogonal with the same span (indeed $\operatorname{span}\{v_1, \dots, v_k\} = \operatorname{span}\{w_1, \dots, w_k\}$ for each k); normalizing gives an orthonormal basis. Every finite-dimensional inner product space has an orthonormal basis.

8.2 Orthogonal complements and projections

- For a finite-dimensional subspace W: $V = W \oplus W^\perp$, and $(W^\perp)^\perp = W$ when V is finite-dimensional.
- Each y is uniquely $u + z$ with $u \in W$, $z \in W^\perp$; if $\{v_i\}$ is an orthonormal basis of W, then $u = \sum \langle y, v_i \rangle v_i$.
- *Best approximation*: u is the unique vector in W closest to y, i.e. $\|y - u\| < \|y - x\|$ for all $x \in W$, $x \neq u$.
- If V is finite-dimensional, $\dim W + \dim W^\perp = \dim V$.

Least squares. For $A \in \mathsf{M}_{m\times n}(\mathbb{F})$ and $y \in \mathbb{F}^m$, the vectors x_0 minimizing $\|Ax - y\|$ are exactly the solutions of the *normal equations* $A^*Ax_0 = A^*y$. If $\operatorname{rank} A = n$, then A^*A is invertible and $x_0 = (A^*A)^{-1}A^*y$.

8.3 The adjoint

Theorem 8.4 (Riesz representation and adjoints). *If V is finite-dimensional and $g \in V^*$, there is a unique $y \in V$ with $g(x) = \langle x, y \rangle$ for all x. Consequently every $\mathsf{T} \in \mathcal{L}(V)$ has a unique* adjoint T^* *with $\langle \mathsf{T}x, y \rangle = \langle x, \mathsf{T}^*y \rangle$, and for an orthonormal basis β, $[\mathsf{T}^*]_\beta = ([\mathsf{T}]_\beta)^*$.*

Rules: $(\mathsf{T} + \mathsf{U})^* = \mathsf{T}^* + \mathsf{U}^*$, $(c\mathsf{T})^* = \bar{c}\,\mathsf{T}^*$, $(\mathsf{TU})^* = \mathsf{U}^*\mathsf{T}^*$, $\mathsf{T}^{**} = \mathsf{T}$, $\mathsf{I}^* = \mathsf{I}$; $\mathsf{N}(\mathsf{T}^*) = \mathsf{R}(\mathsf{T})^\perp$ and $\operatorname{rank}\mathsf{T}^* = \operatorname{rank}\mathsf{T}$.

9 Normal, Self-Adjoint, and Unitary Operators

Definition 9.1. On a finite-dimensional inner product space, T is *normal* if $\mathsf{TT}^* = \mathsf{T}^*\mathsf{T}$; *self-adjoint* (Hermitian) if $\mathsf{T} = \mathsf{T}^*$; *unitary* ($\mathbb{F} = \mathbb{C}$) or *orthogonal* ($\mathbb{F} = \mathbb{R}$) if $\mathsf{TT}^* = \mathsf{T}^*\mathsf{T} = \mathsf{I}$, i.e. $\|\mathsf{T}x\| = \|x\|$ for all x. A matrix A is unitary iff its columns form an orthonormal basis of $\mathbb{F}^n$, iff $A^{-1} = A^*$.

Theorem 9.2 (Schur). *If the characteristic polynomial of T splits, there is an orthonormal basis β for which $[\mathsf{T}]_\beta$ is upper triangular. In matrix form: $A = PUP^*$ with P unitary (orthogonal) and U upper triangular.*

Theorem 9.3 (spectral theorems). Complex: T *is normal* $\iff$ *V has an orthonormal basis of eigenvectors of T.*
Real: T *is self-adjoint* $\iff$ *V has an orthonormal basis of eigenvectors of T.*
In either case, with orthogonal projections T_i onto the eigenspaces $W_i = E_{\lambda_i}$,

$$\mathsf{T} = \lambda_1\mathsf{T}_1 + \cdots + \lambda_k\mathsf{T}_k, \qquad \mathsf{I} = \mathsf{T}_1 + \cdots + \mathsf{T}_k, \qquad \mathsf{T}_i\mathsf{T}_j = \delta_{ij}\mathsf{T}_i, \qquad V = W_1 \oplus \cdots \oplus W_k.$$

- Eigenvalues of a self-adjoint operator are real; those of a unitary operator have absolute value 1.
- If T is normal, $\|\mathsf{T}x\| = \|\mathsf{T}^*x\|$, and $\mathsf{T}x = \lambda x$ implies $\mathsf{T}^*x = \bar{\lambda}x$; eigenvectors for distinct eigenvalues are orthogonal.
- A real symmetric matrix is orthogonally diagonalizable: $A = QDQ^t$ with Q orthogonal and D real diagonal.

9.1 Positive operators and quadratic forms

T is *positive semidefinite* if $\mathsf{T} = \mathsf{T}^*$ and $\langle \mathsf{T}x, x\rangle \geq 0$ for all x, and *positive definite* if moreover $\langle \mathsf{T}x, x\rangle > 0$ for $x \neq 0$; equivalently, T is self-adjoint with nonnegative (positive) eigenvalues. Every positive semidefinite T has a unique positive semidefinite square root, and A^*A is always positive semidefinite. A real quadratic form $q(x) = x^tAx$ with A symmetric is positive definite iff all eigenvalues of A are positive; by Sylvester's law of inertia, the numbers of positive, negative, and zero eigenvalues are invariants under congruence $A \mapsto Q^tAQ$.

10 The Singular Value Decomposition

Theorem 10.1 (SVD)**.** *Let $A \in \mathsf{M}_{m\times n}(\mathbb{F})$ have rank r. There are unitary (orthogonal) $U \in \mathsf{M}_{m\times m}$, $V \in \mathsf{M}_{n\times n}$ and $\Sigma \in \mathsf{M}_{m\times n}$ with $\Sigma_{ii} = \sigma_i$ for $i \leq r$ and all other entries zero, such that*

$$A = U\Sigma V^*, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$

The singular values *σ_i are the square roots of the nonzero eigenvalues of A^*A; the columns of V (right singular vectors) are an orthonormal eigenbasis of A^*A, and $u_i = \sigma_i^{-1}Av_i$ for $i \leq r$.*

- Geometrically, L_A maps the unit sphere to an ellipsoid with semi-axes $\sigma_i u_i$: rotate (V^*), stretch (Σ), rotate (U).
- $\{v_{r+1}, \ldots, v_n\}$ is an orthonormal basis of $\mathsf{N}(L_A)$ and $\{u_1, \ldots, u_r\}$ one of $\mathsf{R}(L_A)$.
- *Pseudoinverse*: $A^\dagger = V\Sigma^\dagger U^*$, where $\Sigma^\dagger$ is $n \times m$ with entries $1/\sigma_i$. Then $x = A^\dagger b$ is the least-squares solution of $Ax = b$ of minimal norm.
- *Polar decomposition*: every square A factors as $A = WP$ with W unitary and $P = (A^*A)^{1/2}$ positive semidefinite.

11 Quick Reference

Dimension theorem	$\operatorname{nullity} \mathsf{T} + \operatorname{rank} \mathsf{T} = \dim V$
Matrix of a map	column j of $[\mathsf{T}]_\beta^\gamma$ is $[\mathsf{T}v_j]_\gamma$
Change of basis	$[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_\beta Q,\ \ Q = [\mathsf{I}]_{\beta'}^{\beta}$
Determinant	multilinear, alternating, $\det I = 1$; $\det(AB) = \det A \det B$
Diagonalizable	f splits and geometric = algebraic multiplicity; p has distinct roots
Cayley–Hamilton	$f(\mathsf{T}) = 0$; $p \mid f$
Jordan form	$V = \bigoplus K_\lambda$; #blocks for $\lambda = \dim E_\lambda$
Gram–Schmidt	$v_k = w_k - \sum_{j<k} \frac{\langle w_k, v_j\rangle}{\|v_j\|^2} v_j$
Projection	$\operatorname{proj}_W y = \sum \langle y, v_i\rangle v_i$ for orthonormal $\{v_i\}$
Least squares	$A^*Ax = A^*y$
Adjoint	$\langle \mathsf{T}x, y\rangle = \langle x, \mathsf{T}^*y\rangle,\ \ [\mathsf{T}^*]_\beta = [\mathsf{T}]_\beta^*$ (β orthonormal)
Spectral theorem	normal ($\mathbb{C}$) / self-adjoint ($\mathbb{R}$) $\iff$ orthonormal eigenbasis
SVD	$A = U\Sigma V^*,\ \ \sigma_i = \sqrt{\lambda_i(A^*A)}$

Reference: S. H. Friedberg, A. J. Insel, L. E. Spence, *Linear Algebra*, 5th ed., Pearson, 2018.